Database Systems Lecture 1: Relational Foundations

Relational Algebra and Schema Normalization

The relational model structures data into mathematical relations represented as tables of tuples and typed attributes
Primary keys uniquely identify relation tuples while foreign key constraints enforce referential integrity.
Third normal form 3NF eliminates transitive functional dependencies to prevent insertion and update anomalies.

Database Systems Lecture 2: Storage and Indexing

B-Tree Indexing and Query Execution Plans

Database storage engines utilize balanced B+ Trees to minimize high-latency disk I/O operations for range queries.
Leaf nodes in a B+ Tree are linked sequentially in a doubly-linked list enabling rapid sequential scans.
Cost-based query optimizers inspect table statistics to select between index seeks, index scans, and hash joins.

Database Systems Lecture 3: Transaction Processing

ACID Guarantees and Two-Phase Locking

A database transaction is an atomic sequence of read and write operations preserving database consistency invaria
The ACID properties define atomicity, consistency, isolation, and durability for mission-critical enterprise systems.
Strict two-phase locking 2PL prevents conflicting concurrent reads and writes, preventing dirty and non-repeatable r